Q-SENSE Error-Mitigation Explorations: Methods, Results, and Reproducible Handoff

12 September 2026

Abstract

This document records the error-mitigation studies performed for the Q-SENSE subspace estimator in this repository. The main test case was H_2O at $R = 1.0$ Å with the optimized variational-orbital (VO) basis and the `FakeQuebec` noise model. We isolated the dominant matrix-entry error, verified noiseless finite-shot statistics, tested exact-symmetry control variates, introduced an exact fixed-orbital sector split, implemented local readout mitigation, and studied deterministic and probabilistic local-folding zero-noise extrapolation (ZNE). We also applied readout mitigation and linear ZNE to the complete 37-state perturbative (PT) matrix. The principal findings are: (i) the original VO energy error is dominated by H_{00}; (ii) splitting the nearly fixed Z_0 sector reduces that dominant entry bias by about a factor of six without approximating the state span; (iii) local readout mitigation is helpful but does not remove the dominant simulator bias; (iv) curved ZNE fits reduce bias for one deep split-basis diagonal entry, at substantial variance cost; and (v) applying the original Q-SENSE shot allocation naively to every entry and ZNE factor is unreliable for the larger PT matrix. Numerical claims below are tied to saved artifacts so they can be checked independently.

1 Scope and status

All noisy results in this note are local finite-shot Qiskit Aer simulations. They use the calibration data, connectivity, native gate target, and noise model supplied by `FakeQuebec`; they are not executions on an IBM quantum processor. The problem circuits use Aer's `statevector` method with `enable_truncation=True`. Consequently these are noisy stochastic, shot-based statevector simulations rather than density-matrix calculations. For the wide but unentangled readout-calibration circuits used by the full-matrix scripts, the method was changed to `matrix_product_state` to avoid allocating a 127-qubit statevector.

The baseline Q-SENSE accuracy request is

$$\epsilon_E = 1.6 \times 10^{-3} \text{ Ha}.$$

This value is set by `TARGET_ESTIMATE_ACCURACY = 1.6e-3` in `scripts/sampling_and_sv.py`. The original and split VO simulations predate the portable-output default and therefore retain several trusted local pickle files. Newer mitigation runs save numerical data in JSON, NPZ, CSV, and circuits in QPY wherever practical.

1.1 Terminology and error metrics

For a matrix entry, bias means the observed estimate minus its exact noiseless value,

$$b_{ij} = \widehat{H}_{ij} - H_{ij}^{\text{ideal}}.$$

For an energy estimate, the same sign convention is used. Standard deviations reported by the estimator describe finite-shot uncertainty under the sampled distribution; they do not include uncertainty in the noise model or, unless explicitly stated, the readout-calibration matrices. We use

$$\mathrm{RMSE} = \sqrt{b^2 + s^2}$$

when s^2 is the estimated sampling variance. Whenever a noisy matrix develops spurious eigenvalues below the desired state, both the algebraically lowest eigenpair and the eigenpair with maximum overlap with the ideal coefficient vector are reported. The latter is called the *overlap-tracked target*; it is a simulator diagnostic because the ideal vector would not be known on hardware.

2 Baseline Q-SENSE estimation

Let $\{|\Phi_i\rangle\}$ be an orthonormal Q-SENSE basis and

$$H_{ij} = \langle \Phi_i | \hat{H} | \Phi_j \rangle, \qquad \mathbf{H}\vec{c} = E\vec{c}.$$

Each quantum matrix element is decomposed into jointly measurable fragments f. If $\sigma_{ij,f}$ is the exact single-shot standard deviation of a fragment, the ideal first-order energy sensitivity gives the weights

$$w_{ij} = \begin{cases} |c_i|^2 \sum_f \sigma_{ii,f}, & i = j, \\ 2|c_i c_j| \sum_f \sigma_{ij,f}, & i > j. \end{cases}$$

Writing $M = \sum_{i \geq j} w_{ij}$, the implemented total-shot estimate and fragment allocation are, up to ceiling,

$$N = \left\lceil \frac{M^2}{\epsilon_E^2} \right\rceil, \tag{1}$$

$$N_{ij,f} = \left\lceil N \frac{w_{ij}}{M} \frac{\sigma_{ij,f}}{\sum_g \sigma_{ij,g}} \right\rceil. \tag{2}$$

Thus observable coefficients matter through the exact fragment standard deviations and the shot allocation. They are not encoded in a PUB's `target_precision`: the local `BackendEstimatorV2` chooses its number of shots as $\lceil 1/\epsilon_{\mathrm{PUB}}^2 \rceil$, so the code passes

$$\epsilon_{\mathrm{PUB}} = \frac{1 + 10^{-12}}{\sqrt{N_{ij,f}}}$$

only to reproduce the already calculated shot count exactly.

Before execution, state-preparation and interference circuits are transpiled against the `FakeQuebec` target with optimization level 3 and transpiler seed 0. This enforces its native gate set and limited connectivity. The nominal circuit width remains 127 in saved QPY objects, but Aer truncation simulates only active connected qubits; the completed VO runs reported ranges of 5–19 active qubits across groups. A local `CapturingBackendEstimatorV2` is used directly, without a Runtime batch, and stores the exact measurement histograms used for its expectations.

2.1 Original 11-state VO baseline

The completed run is

```
saved/runtime_runs/
  20260722T164209Z_h2o_1.0_qwc_fake_quebec/
```

It used 45 quantum matrix elements, 585 PUBs, and 205,731 retained shots. The maximum transpiled circuit depth was 1006 and the maximum two-qubit gate count was 376. Runtime-level mitigation was disabled. The result was

$$E_{\text{ideal}} = -75.0176664101 \text{ Ha}, \qquad \widehat{E} = -74.2180687901 \text{ Ha},$$

with

$$\text{bias} = +0.7995976200 \text{ Ha}, \qquad \text{standard deviation} = 0.7643728997 \text{ Ha}, \qquad \text{RMSE} = 1.1061746172 \text{ Ha}.$$

The leading ideal coefficient weight is $|c_0|^2 = 0.9846882$. The corresponding diagonal entry had

$$H_{00}^{\text{ideal}} = -74.9897246858, \qquad \widehat{H}_{00} = -59.9089790839,$$

or bias +15.0807456019 Ha. Replacing only this noisy entry by its ideal value while retaining all other noisy entries gives an energy bias of +0.0275449236 Ha and coefficient-vector fidelity 0.9839845. This counterfactual establishes that H_{00}, rather than the accumulation of all other errors, dominated this particular run.

2.2 Noiseless check of H_{00} statistics

A finite-shot run with no noise model gave entry bias +0.00246196 Ha. A separate 10,000-replicate bootstrap found

quantity	standard error (Ha)
analytic Q-SENSE prediction	0.00139248
parametric bootstrap from exact probabilities	0.00138930
nonparametric bootstrap of the finite histogram	0.00156161

The parametric result agrees with the analytic calculation to 0.23%. The larger nonparametric value reflects the particular finite histogram and rare outcomes. The saved quantity 0.21810137 for fragment 0 is a *single-shot* standard deviation, not the standard error after 60,882 shots. These checks support the noiseless variance implementation and show that the large noisy bias is not a failure of ideal shot-noise accounting.

3 Exact observable shifts as control variates

Suppose the target state obeys an exact constraint $\langle C \rangle = c$. The shifted estimator

$$\widehat{O}_\lambda = \widehat{O} + \lambda(\widehat{C} - c)$$

has the same ideal expectation as O. When O and C are evaluated from the same bit strings,

$$\operatorname{Var}(\widehat{O}_\lambda) = \operatorname{Var}(\widehat{O}) + 2\lambda \operatorname{Cov}(\widehat{O}, \widehat{C}) + \lambda^2 \operatorname{Var}(\widehat{C}),$$

so the variance-optimal coefficient is

$$\lambda_{\text{var}} = -\frac{\operatorname{Cov}(\widehat{O}, \widehat{C})}{\operatorname{Var}(\widehat{C})}.$$

This coefficient minimizes variance, not hardware bias. A coefficient

$$\lambda_{\text{bias}} = -\frac{\widehat{O} - O_{\text{ideal}}}{\widehat{C} - c}$$

was also computed to diagnose whether a control could cancel the observed bias. It is labelled *diagnostic bias-optimal* because it uses the ideal answer and must not be selected this way in an unknown production problem.

All results in this section use entry $(0, 0)$, fragment 0, its 60,882 saved shots, and 20,000 multinomial bootstrap resamples. The ideal fragment value is -30.6034385637 Ha. The raw bootstrap fragment bias is approximately $+14.70$ Ha; the other fragments and exact classical contribution make the raw full-entry bias approximately $+14.80$ Ha.

3.1 Global parity

The first control was the tapered-register parity $P = \prod_q Z_q$ with eigenvalue $p = -1$. With $\lambda = -24.8194965$, chosen from the largest observable coefficient, the sampled parity was -0.1790020. The full-entry bias changed from $+14.8021$ to -5.5732 Ha, while the bootstrap standard deviation increased from 0.08538 to 0.12087 Ha. The correction therefore crossed the ideal value: the large parity violation multiplied by a coefficient chosen for Pauli-coefficient cancellation rather than noise-response calibration. The variance-optimal coefficient, -3.2584, barely improved the bias ($+12.0240$ Ha at fragment level). The diagnostic bias-optimal coefficient was -17.9045.

3.2 Pair number

For seven tapered occupation qubits, the pair-number operator is

$$N_{\mathrm{pair}} = \frac{7}{2} I - \frac{1}{2} \sum_{q=0}^{6} Z_q, \qquad \langle N_{\mathrm{pair}} \rangle = 5.$$

The same noisy bit strings gave $\widehat{N}_{\mathrm{pair}} = 4.5888768$, much closer to the exact value than the parity estimator was to -1. Let $a = 24.8194965$ denote the magnitude used in the coefficient arguments. The tested results were

choice	λ	full-entry bias	bootstrap std.	RMSE
raw	0	14.8014	0.08561	14.8017
$2a/5$	9.9278	10.7202	0.07320	10.7204
$2a/3$	16.5463	7.9993	0.07723	7.9997
$4a/3$	33.0927	1.1972	0.11918	1.2031
λ_{var}	10.6252	10.4335	0.07314	10.4337

The variance-optimal full-entry values were reconstructed from the saved fragment samples by adding the same fixed contribution as in the other rows. The diagnostic bias-optimal value was 35.7547. The closeness of $2a/5$ to λ_{var} is empirical, not a general identity: $2a/5$ follows a coefficient-balancing argument, whereas λ_{var} follows a sampled covariance ratio.

3.3 Z_0-weighted number control

To make the control share more of the dominant Z_0 term's noise response, we tested

$$C = (N_{\mathrm{pair}} - 5) Z_0, \qquad \langle C \rangle = 0.$$

For seven qubits,

$$C = -\frac{1}{2} \left(3Z_0 + I + \sum_{j \neq 0} Z_0 Z_j \right).$$

This algebra explains why $\lambda = 2a/3$ cancels the explicit dominant Z_0 coefficient in the shifted observable. Cancellation of that one coefficient does *not* force the noisy expectation of the entire shifted quantum part to vanish, because the identity and six Z_0Z_j terms remain. The measured control value was -0.1876745.

The variance-optimal value $\lambda = 10.2948$ reduced the standard deviation from 0.08559 to 0.07231 Ha but left full-entry bias +12.8710 Ha. Coefficient cancellation ($\lambda = 16.5463$) left bias +11.6976 Ha. The deliberately large choice $\lambda = 2a = 49.6390$ reduced full-entry bias to +5.4862 Ha while increasing the standard deviation to 0.18940 Ha. The diagnostic bias-optimal coefficient 78.3250 produced bias +0.1019 Ha and standard deviation 0.31120 Ha, but it is target-leaking and therefore not a usable mitigation prescription.

The control's shared Z_0 and Z_0Z_j support gives it a more similar noise profile to the dominant observable term than $N_{\text{pair}} - 5$ alone. This is a plausible mechanism, not a theorem under the gate-dependent `FakeQuebec` model. In the actual diagnostic, the unweighted number control with $\lambda = 4a/3$ gave a better operationally motivated bias result than the tested weighted coefficients. A future control should be selected from independent pilot data or a noise model using covariance and response similarity, then frozen before evaluating production data.

4 Exact fixed-orbital sector split

The dominant VO state was found to occupy tapered orbital/qubit 0 almost entirely in $|1\rangle$. Rather than use Z_0 only as a noisy measured control, state 0 was split into its normalized fixed-Z_0 sectors. Six pair generators acting on orbital 0 were removed; the remaining 24 generators were warm-started from the original angles and independently reoptimized in each six-qubit active sector. The results were

	$q_0 = 0$	$q_0 = 1$
sector probability	5.1370×10^{-7}	0.9999994863
number of determinants	6	15
optimized infidelity	6.0×10^{-15}	1.7×10^{-13}
active-circuit two-qubit gates	48	48

Their weighted coherent reconstruction has fidelity 0.9999999999998239 with the original 21-determinant state. In the expanded Q-SENSE basis the two normalized sectors are used as separate states, without their reconstruction weights. They are mutually orthogonal by construction, and the remaining old basis states lie in different pre-existing symmetry sectors as required by the method. No generalized eigenvalue problem is introduced. Qubit 0 is classical and removed from the active state circuits for these two states.

The split basis contains 12 rather than 11 states. Its dominant physical component is entry $(1, 1)$, corresponding to the $q_0 = 1$ sector, with $|c_1|^2 = 0.9846877$. Its raw entry bias was +2.5567470 Ha, versus +15.0807456 Ha for the unsplit H_{00}. If only split entry $(1, 1)$ is set to ideal, the remaining noisy matrix gives energy bias +0.0228070 Ha and coefficient fidelity 0.979334. This confirms a large structural bias reduction for the dominant entry.

The single complete split run nevertheless had energy bias +0.8839538 Ha and coefficient fidelity 1.73×10^{-5} because the biased dominant diagonal no longer remained the lowest state in the noisy matrix. It used 51 quantum matrix elements, 626 PUBs, and 201,422 retained shots. This result does not negate the entry-level improvement; it shows that the total estimator still needs mitigation or sufficient sampling for the other entries.

5 Local readout-error mitigation

5.1 Implemented estimator

For every measured physical qubit q, two calibration experiments estimate

$$A_q = \begin{pmatrix} p(0|0) & p(0|1) \\ p(1|0) & p(1|1) \end{pmatrix}.$$

All support qubits are prepared and measured in parallel, so the independent local model requires only two calibration circuits: one all-zero and one all-one circuit, each with S_{cal} shots. For a measured Pauli product with support Q, the corrected outcome weight is evaluated using

$$\bigotimes_{q \in Q} A_q^{-1}.$$

The code applies these factors directly to each observed bit string and computes both the corrected first and second moments. It therefore avoids constructing a dense $2^{|Q|} \times 2^{|Q|}$ matrix. Calibration cost is $2S_{\mathrm{cal}}$ shots, independent of support width under the parallel local model, while per-outcome processing is linear in Pauli support.

The statistical overhead is not width-independent. Defining the readout contrast

$$r_q = 1 - p(1|0) - p(0|1),$$

a symmetric-error approximation gives a worst-case variance amplification for a Pauli product of roughly

$$\gamma_Q \simeq \prod_{q \in Q} r_q^{-2}.$$

Thus local inversion is computationally scalable but can become statistically expensive for long Pauli strings or poorly conditioned A_q.

The preferred split-diagonal calibration used $S_{\mathrm{cal}} = 20{,}000$ and physical qubits $\{45, 54, 62, 63, 64, 65\}$. The assignment matrices were nearly symmetric, with single-qubit misassignment probabilities between about 0.42% and 1.90%. For example,

$$A_{62} = \begin{pmatrix} 0.98105 & 0.01870 \\ 0.01895 & 0.98130 \end{pmatrix}.$$

This symmetry is why readout twirling was not expected to add a large benefit; readout twirling itself has not been tested.

5.2 Split-basis diagonal results

The paired comparison reran the same circuits and reports raw and mitigated estimates from the same histograms. This is the correct comparison; the "previous bias" column came from an independent earlier run and can differ because of finite shots. For the important $(1, 1)$ entry,

$$b_{11}^{\text{paired raw}} = +2.577674 \text{ Ha}, \qquad b_{11}^{\text{readout}} = +2.498453 \text{ Ha}.$$

Readout inversion removed only 0.07922 Ha of a roughly 2.58 Ha bias. Across the six quantumly sampled diagonals, the absolute paired bias decreased for all six; the mean absolute bias fell from 1.6487 to 1.3961 Ha. Several of these entries had only 3–317 shots, so individual improvements and zero empirical standard deviations are noisy diagnostics, not precision benchmarks. Fully classical entries are copied exactly and assigned zero bias and variance by construction.

5.3 Unpropagated calibration uncertainty

Current error bars condition on the estimated A_q and omit its finite-shot uncertainty. A complete analysis should resample each calibration histogram, rebuild every A_q^{-1}, resample problem histograms, and repeat mitigation, matrix assembly, and diagonalization. This hierarchical bootstrap directly provides the calibration-induced component of entry and energy variance. Choose S_{cal} so that this component is a small predeclared fraction of the total variance budget. The calibration counts are saved, so no new noisy run is required to implement this analysis.

The reusable code is in `src/mitigation.py`; the main switch is `enable_readout_mitigation` or environment variable `QSENSE_ENABLE_READOUT_MITIGATION`. Runtime resilience levels are not used: the mitigation stack is explicit.

6 Zero-noise extrapolation

6.1 Folding and transpilation convention

For an eligible native gate U, local folding replaces

$$U \longmapsto UU^{\dagger}U,$$

which is ideal-equivalent but executes additional noisy operations. Full local folding was first tested at factors $\{1, 3, 5\}$. For a fractional factor $1 \leq \lambda \leq 3$, each eligible gate is independently folded with probability

$$p_{\mathrm{fold}} = \frac{\lambda - 1}{2}.$$

Eight randomized folding patterns are used for each non-unit factor. The factor's fixed shot budget is divided among the patterns; randomization raises the circuit/PUB count, not the total shots, unless the user separately raises the per-factor budget.

The source circuits have already been transpiled at optimization level 3 to the `FakeQuebec` native basis and layout. Folding is then performed in that basis. Only operations whose backend target reports nonzero error are eligible. IBM-style R_z operations are normally virtual frame changes and the fake backend reports zero error, so they are not folded. This also typically holds on a real IBM device when the final ISA circuit retains an R_z, but the live backend target must be checked. The inverse of an `sx` is implemented in the native basis as $R_z(\pi)\,\mathtt{sx}\,R_z(-\pi)$. If folding introduces an unsupported gate, it is transpiled at optimization level 0 with the original layout to avoid optimizing away the fold.

The estimator constructs measurement-basis rotations after this folding stage. Those rotations are not folded in the current implementation. Therefore λ scales the saved state/interference circuit's eligible gates, not every error source in the complete measured circuit. Realized gate counts and fold seeds are saved and should be checked.

In an ideal noiseless calculation all folding patterns implement the same unitary, so their exact expectation values are identical. Under gate-dependent and location-dependent noise, patterns with the same expected λ can have different means. Finite-shot noise can mask this between-pattern variance; the implementation combines resolved pattern spread with within-pattern shot variance.

6.2 Linear and variance-weighted extrapolation

For factors λ_i, ordinary linear ZNE fits

$$E(\lambda) = \beta_0 + \beta_1 \lambda,$$

and takes $\widehat{E}(0) = \widehat{\beta}_0 = \sum_i c_i \widehat{E}_i$. Variance-weighted linear ZNE minimizes

$$\chi^2 = \sum_i \frac{[\widehat{E}_i - (\beta_0 + \beta_1 \lambda_i)]^2}{v_i},$$

using inverse total factor variances. For independent factors,

$$\mathrm{Var}[\widehat{E}(0)] = \sum_i c_i^2 v_i.$$

If the single-shot standard deviations are σ_i and the coefficients are fixed, the variance-optimal allocation at fixed total shots is

$$N_i \propto |c_i| \sigma_i.$$

There is a design loop: changing N_i changes v_i, which changes the weighted-fit coefficients c_i. The consistent practical solution is to estimate σ_i and weights in an independent pilot, freeze them, impose a minimum diagnostic allocation at interior factors, and then execute the production allocation once. Repeatedly adapting weights to the same data would invalidate a simple uncertainty interpretation.

6.3 Full-fold diagnostic on split entry $(0, 0)$

The initial $\{1, 3, 5\}$ full-fold test concerned the low-weight split entry $(0, 0)$, not the dominant $(1, 1)$ entry. With 60,000 shots per factor and readout mitigation, the factor estimates were

$$(-27.66718, -23.69699, -21.66888) \text{ Ha}.$$

Linear extrapolation gave -28.84307 Ha versus ideal -31.03017 Ha: bias $+2.18710$ Ha, empirical standard deviation 0.22192 Ha, and RMSE 2.19833 Ha. The scale-one bias was $+3.36299$ Ha. This result is distinct from the earlier $2.5777 \to 2.4985$ Ha readout result, which concerned split entry $(1, 1)$ and did not apply ZNE.

Factors 3 and 5 were judged unnecessarily severe for these already deep, noisy circuits because amplified values moved toward saturation. Subsequent work used smaller probabilistic factors.

6.4 Probabilistic ZNE on dominant split entry $(1, 1)$

With factors $\{1, 1.5, 2\}$, eight patterns per non-unit factor, and the cached per-factor fragment allocation, the scale-one and linear-ZNE results were

	bias (Ha)	std. (Ha)	RMSE (Ha)
$\lambda = 1$	2.49492	0.06567	2.49579
ordinary linear ZNE	1.30222	0.14826	1.31063

The extrapolation reduced bias by about 48% but more than doubled the standard deviation.

A pilot-based allocation preserved the same total of 441,951 shots but used

$$(N_1, N_{1.5}, N_2) = (234371, 45356, 162224).$$

The fitted intercept coefficients became

$$(c_1, c_{1.5}, c_2) = (1.98889, 0.02221, -1.01111).$$

The resulting bias, standard deviation, and RMSE were 1.21341, 0.12433, and 1.21976 Ha, respectively. Relative to the equal-allocation run, the variance decreased by 29.7% and RMSE by 6.9%.

6.5 Curved models and five-factor study

The final grid was

$$\lambda \in \{1, 1.25, 1.5, 1.75, 2\}.$$

Two independently seeded runs were inverse-variance pooled, for a nominal 883,902 shots per factor. The pooled data were

λ	$\widehat{E}(\lambda)$	standard deviation
1.00	−72.509300	0.02681
1.25	−72.133633	0.02773
1.50	−71.774107	0.03487
1.75	−71.453188	0.02890
2.00	−71.160255	0.03185

Three variance-weighted extrapolators were compared:

$$E_{\text{lin}}(\lambda) = \beta_0 + \beta_1 \lambda, \tag{3}$$

$$E_{\text{quad}}(\lambda) = \beta_0 + \beta_1 \lambda + \beta_2 \lambda^2, \tag{4}$$

$$E_{\text{exp}}(\lambda) = E_0 + s[1 - e^{-k\lambda}]/k. \tag{5}$$

Factor-level parametric bootstrap with 10,000 samples gave

model	$E(0)$	bias	std.	RMSE
linear	−73.84512	1.14458	0.0554	1.1459
quadratic	−74.32102	0.66868	0.2752	0.7231
offset exponential	−74.40536	0.58434	0.3809	0.6975

Curved models reduced observed simulator bias but amplified variance. The quadratic coefficient was $\beta_2 = -0.2294 \pm 0.1293$ with bootstrap 95% interval $[-0.4849, 0.0254]$ and two-sided $p = 0.076$. Holding out $\lambda = 1.75$, standardized prediction residuals were 0.632, 0.041, and 0.087 for linear, quadratic, and exponential fits. Curvature is therefore supported by the trend and prediction test but was not conventionally significant at $p < 0.05$.

For Gaussian residuals, the reported AIC $= \chi^2 + 2k$ penalizes fit residuals by twice the number of model parameters. The AIC values were 7.129, 6.034, and 6.051. Differences below roughly 2 do not decisively select a model, so the quadratic's 1.095 advantage over linear was suggestive rather than conclusive. More independent factors are more valuable for detecting curvature than simply reducing error bars on the same three points, although both help. There is no universal circuit depth at which a curved model becomes correct; accumulated error, observable support, gate composition, and saturation matter together [2, 3].

7 Complete-matrix mitigation experiments

7.1 Original 11-state VO matrix

A script was written to fix $(0, 0)$ at its ideal value and apply local readout mitigation plus probabilistic linear ZNE to every other quantum entry: `scripts/run_full_matrix_readout_linear_zne.py`. The first attempt failed because readout calibration with a 127-qubit statevector exceeded `max_memory_mb`. Calibration was changed to MPS, while problem circuits retained truncated noisy statevector simulation. No complete 11-state full-matrix mitigated result is currently saved. The incomplete directories ending in `20260813T032929Z` and `20260813T033440Z` must not be treated as results.

The useful unmitigated counterfactual is the calculation above: with ideal H_{00} and every other original noisy entry unchanged, the energy bias is +0.0275449 Ha. This is the baseline a future completed full-matrix mitigation run should beat.

7.2 Completed 37-state PT matrix

The analogous PT workflow completed and is stored under

```
saved/runtime_runs/pt_h2o_1.0/
  pt_readout_linear_zne_20260813T134200Z/
```

It fixed $(0, 0)$ to ideal, used factors $\{1, 1.5, 2\}$, eight patterns per non-unit factor, and respective totals of 19,900, 33,937, and 33,937 shots. There were 40,358 folded PUBs in 17 sequential simulator batches. The run used the original Q-SENSE allocation independently at each factor.

matrix	lowest bias	lowest overlap	tracked bias	tracked overlap
same-shot raw	−8.35454	0.000781	−0.005163	0.91069
readout only	−9.17104	0.000777	−0.068029	0.71424
readout + linear ZNE	−34.28791	0.000437	−0.113953	0.80240

Large noisy diagonal errors create spurious low eigenvalues, so the lowest energy is not a meaningful estimate here. Even for the overlap-tracked state, readout correction and linear ZNE worsened energy bias relative to a fortuitously accurate raw result. This is evidence against applying the baseline allocation mechanically to all entries and factors; it is not evidence that ZNE must fail after a redesigned allocation.

Many fragments received one or only a few shots. A one-shot empirical variance is zero but does not mean the estimator is exact. Post-processing therefore used the floor

$$v_{ij}^{\text{used}} = \max\left(v_{ij}^{\text{empirical}}, \sum_f \frac{\sigma_{ij,f,\text{ideal}}^2}{N_{ij,f}}\right).$$

This floor was applied 506 times. Forty-seven ill-conditioned weighted linear fits fell back to ordinary least-squares coefficients while retaining the same linear model. These counts diagnose an underresolved design.

7.3 Leading PT fragment and increased-shot test

Entry $(23, 0)$, fragment 0 was isolated as a leading cause of coefficient vector error. Its ideal entry is 2.3553×10^{-5} Ha, but the original four-shot raw fragment estimate was −0.17943 Ha. The transpiled fragment circuit has depth 254 and 90 native ECR gates. The two PT states contain only two and one pair rotations, with angles -6.99×10^{-4} and -4.78×10^{-4} radians; their individual logical preparations have depths 15 and 8 with four and three CX gates. Hence most of the 90-ECR cost comes from the interference/controlled-swap construction and routing, not from the small pair rotations.

Eight independent noisy replicas at 100, 500, 2,000, and 10,000 shots showed that more shots reduce spread but do not remove device-model bias. At 10,000 shots the mean entry biases were +0.03780 Ha raw and +0.04215 Ha after readout correction. The corresponding overlap-tracked energy biases were +0.037685 and +0.032139 Ha. Thus the original four-shot value happened to cancel other matrix errors and gave a misleadingly small total-energy bias. In a separate noise-free study with 32 replicas per shot count, the mean bias remained statistically consistent with zero and the single-run standard deviation decreased from 3.03×10^{-4} at 100 shots to 3.36×10^{-5} at 10,000 shots, consistent with $N^{-1/2}$ convergence.

8 Conclusions supported by the data

1. The original H_2O VO error at $R = 1.0$ Å is overwhelmingly tied to one high-weight diagonal entry. Increasing shots cannot remove its noise-model bias.
2. Exact control variates can trade bias against variance when the control experiences correlated noise. Variance-optimal and bias-optimal coefficients answer different questions. Coefficients fitted using the ideal target are diagnostics only.
3. The fixed-Z_0 sector split is the strongest presently demonstrated operational improvement to the dominant entry. It changes the basis and measurement cost but preserves the original state exactly within numerical optimization error.
4. Local readout inversion consistently improved the paired sampled diagonals, but the dominant-entry improvement was modest. Readout is therefore not the main error source under this fake-backend model.
5. Probabilistic ZNE with small factors is preferable here to full factors 3 and 5. For split entry $(1, 1)$, ZNE roughly halved bias. Curved models reduced RMSE further, but with much larger uncertainty and no decisive model-selection result.
6. The completed PT matrix shows that entrywise mitigation can destabilize diagonalization when shots are allocated for the unmitigated first-order energy objective. Full-matrix mitigation needs a new joint allocation accounting for extrapolation coefficients, readout overhead, minimum per-fragment resolution, and eigenvalue sensitivity.

9 Known limitations and recommended next work

1. Implement a hierarchical bootstrap that resamples calibration and problem histograms together. Use it to choose S_{cal} and to propagate uncertainty through diagonalization.
2. Redesign full-matrix allocation using pilot estimates of each factor's variance and fixed extrapolation coefficients. Enforce a scientifically chosen minimum number of shots per retained fragment; never interpret one-shot empirical variance as zero uncertainty.
3. Fold the complete physical measurement circuit, including estimator basis rotations, or explicitly measure the unamplified contribution.
4. Test model stability under modest changes of factor grid, fold seed, and held-out factor. Reject extrapolation when amplified values have saturated or the intercept is unstable.
5. Explore additional exact controls that share the dominant Pauli support. Select coefficients from independent pilots or a justified noise model, not proximity to an ideal answer.
6. Generalize the fixed-orbital split only where occupation diagnostics show an almost fixed tapered qubit. Record reconstruction fidelity, enlarged matrix cost, and resulting shot allocation at every geometry.
7. Revalidate the conclusions on current real-device calibration data. `FakeQuebec` establishes behavior of one local model, not a hardware-independent result.

10 Reproduction guide

Run commands from the repository's inner `seniority/` directory using the `SenDev` environment. The principal scripts are:

```
# Build and verify the exact fixed-orbital split
python scripts/build_orbital0_split_states.py

# Baseline split sampling and local noisy simulation
```

```
python scripts/sampling_and_sv_orbital0_split.py

# Readout-mitigated split diagonals
python scripts/run_readout_mitigated_diagonals.py \
  --run-dir saved/runtime_runs/<split-run>

# One-entry probabilistic readout + ZNE
python scripts/run_probabilistic_zne_entry.py --entry 1

# Compare linear, quadratic, and exponential models
python scripts/fit_zne_models.py <result-directories>

# Inspect a full-matrix job without running it
python scripts/run_full_matrix_readout_linear_zne.py --dry-run

# Resume PT aggregation from already completed batches
python scripts/run_pt_readout_linear_zne.py \
  --resume-run saved/runtime_runs/pt_h2o_1.0/\
pt_readout_linear_zne_20260813T134200Z
```

Command-line help should be consulted before a new expensive run because defaults and required paths differ among the exploratory scripts.

Portable JSON can be loaded without reconstructing Qiskit result objects:

```
from pathlib import Path
from src.saved_run_io import load_json
import numpy as np

run = Path("saved/runtime_runs/<run-directory>")
metadata = load_json(run / "run_results.json.gz")
histograms = load_json(run / "raw_measurement_histograms.json.gz")
arrays = np.load(run / "matrices_and_eigenvectors.npz",
                 allow_pickle=False)

ideal_matrix = arrays["ideal_matrix"]
noisy_matrix = arrays["noisy_matrix"]
ideal_c = arrays["ideal_eigenvector"]
noisy_c = arrays["noisy_eigenvector"]
```

For mitigation result directories, the principal files are

```
results.json.gz
fragment_results.json.gz
batches/*.json.gz
```

as indicated by the script README. Load QPY circuits with `qiskit.qpy.load`. Legacy `*.pkl` and `*.dump` files should be unpickled only when they are trusted local artifacts; Qiskit container pickles are version-sensitive and one historical raw `DataBin` pickle is known to be unreadable under a newer Qiskit version.

10.1 Authoritative artifact map

Purpose	Repository-relative path
Original VO baseline	`saved/runtime_runs/20260722T164209Z_h2o_1.0_qwc_fake_quebec/`
Noiseless and symmetry-shift diagnostics	Same directory; files beginning `bootstrap_noiseless`, `noiseless_control`, `symmetry_shift`, `number_shift`, and `z0_weighted_number_shift`.
Split-state construction	`data/h2o_data/UCSF_sym_comp_for_Praveen_Smik_1.0_state0_orbital0_split.dump` and companion `.txt`.
Split VO baseline	`saved/runtime_runs/20260724T024719Z_h2o_1.0_state0_orbital0_split_v1_qwc_fake_quebec/`
Preferred readout diagonal run	Split run plus `readout_mitigated_diagonals_20260724T211017Z/`.
Initial and optimized three-factor ZNE	Split run plus `probabilistic_zne_entry_1_1_20260725T200028Z/` and `...20260725T203218Z/`.
Final five-factor studies	Split run plus `probabilistic_zne_entry_1_1_20260727T161644Z/` and `...20260727T162129Z/`.
Detailed ZNE-only note	`documents/zne_study_summary.tex`.
Completed PT full-matrix run	`saved/runtime_runs/pt_h2o_1.0/pt_readout_linear_zne_20260813T134200Z/`.
Reusable mitigation routines	`src/mitigation.py`.
Raw-count estimator adapter	`src/capturing_backend_estimator.py`.
Script inventory	`scripts/README.md`.

References

[1] T. Giurgica-Tiron, Y. Hindy, R. LaRose, A. Mari, and W. J. Zeng, "Digital zero noise extrapolation for quantum error mitigation," *2020 IEEE International Conference on Quantum Computing and Engineering*, 306–316 (2020). doi:10.1109/QCE49297.2020.00045.

[2] R. Majumdar, P. Rivero, F. Metz, A. Hasan, and D. S. Wang, "Best practices for quantum error mitigation with digital zero-noise extrapolation," *2023 IEEE International Conference on Quantum Computing and Engineering*, 881–887 (2023). doi:10.1109/QCE57702.2023.00102.

[3] Z. Cai, "Multi-exponential error extrapolation and combining error mitigation techniques for NISQ applications," *npj Quantum Information* **7**, 80 (2021). doi:10.1038/s41534-021-00404-3.

[4] M. Krebsbach, B. Trauzettel, and A. Calzona, "Optimization of Richardson extrapolation for quantum error mitigation," arXiv:2201.08080 (2022). https://arxiv.org/abs/2201.08080.